StemSet — Scientific Paper Template User Manual

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Abstract

This document is the official user manual for the **StemSet** scientific paper template. It covers every feature provided by the `stemset.sty` package: paper metadata, abstract and keywords, theorem environments, coloured boxes, STEM and electrical-engineering math macros, code listings, TikZ diagram macros, algorithm pseudocode, smart cross-referencing, acronyms, SI units, and IEEE bibliography. Each feature is explained with ready-to-use examples.

Keywords: LaTeX, template, scientific paper, electronic engineering, STEM

Contents

1 Structure and General Settings

The template is designed to provide a complete environment for writing scientific papers and academic papers in IEEE conference style. Before diving into the details of individual modules, this chapter covers the basic features included in the template and how to configure the main document.

1.1 Acronym Management (acro)

For technical documents, acronyms are vital. The template uses the `acro` package. Acronyms must be defined in the preamble of the main file using the `\DeclareAcronym` command:

```
\DeclareAcronym{fpga}{
  short = FPGA,
  long  = Field Programmable Gate Array
}
```

In the text, you can reference the acronym using:

- `\ac{fpga}`: Prints the full form on first use (e.g. Field Programmable Gate Array (FPGA)) and the short form on subsequent uses (e.g. FPGA).
- `\acl{fpga}`: Always forces the printing of the full form (e.g. Field Programmable Gate Array).
- `\acs{fpga}`: Always forces the printing of the short form (e.g. FPGA).

1.2 Bibliography Usage (biblatex)

The bibliography is managed automatically using `biblatex` with the `biber` backend and the IEEE style. To insert sources:

1. Add your references in BibTeX format to the `layout/bibliografia.bib` file.
2. Cite sources in your text using the `\cite{bibtex_key}` command (e.g. **razavi2001cmos**).
3. Print the bibliography at the end of the document by inserting the `\printbibliography` command.

1.3 Main Module: stemset.sty

The base package `stemset.sty` is the core of the template. It acts as an "Hub" to load all other modules and manages the language and style settings for the entire document.

How to Use (Code Example)

The package is loaded in the preamble of the `main.tex` document. It uses a key-value interface for configuration.

```
\usepackage[
  language=english,      % Document language: 'italian' or 'english'
  modules={math, boxes, code, algorithms, tikz} % Modules to load
]{package/stemset}
```

Available Options

- **`language`**: Automatically sets translations for algorithms, theorems, and layout labels (`italian` or `english`).
- **`modules`**: A comma-separated list of submodules to activate.

2 Boxes and Theorems Module: stemset-boxes.sty

This module manages the aesthetics of all information boxes and mathematical environments (theorems, definitions, etc.) suitable for a two-column layout.

2.1 What it does

It allows you to call structured elements semantically, automatically applying a clean graphical layout in line with scientific publishing standards.

2.2 How to Use (Code Example)

The syntax is: `\nomemacro{Title}{Text}` for elements requiring a title, or `\nomemacro{Text}` for notes and warnings.

```
\definition{Causality}{A system is causal if...}

\note{Ensure that the sampling frequency satisfies the Nyquist-Shannon theorem.}
```

2.3 Full List of Features

Below are all the information and mathematical environments available:

- **Definition**: `\definition{Title}{Text}` | **IT**: `\definizione`
- **Theorem**: `\theorem{Title}{Text}` | **IT**: `\teorema`
- **Corollary**: `\corollary{Title}{Text}` | **IT**: `\corollario`
- **Proposition**: `\proposition{Title}{Text}` | **IT**: `\proposizione`
- **Note**: `\note{Text}` | **IT**: `\nota`
- **Warning**: `\warning{Text}` | **IT**: `\attenzione`
- **Example**: `\example{Title}{Text}` | **IT**: `\esempio`
- **Remember**: `\remember{Text}` | **IT**: `\ricorda`
- **Tip**: `\tip{Text}` | **IT**: `\suggerimento`
- **Exam Question**: `\examquestion{Text}` | **IT**: `\domandaesame`
- **Curiosity**: `\curiosity{Text}` | **IT**: `\curiosita`

2.4 Visual Result

Definition: Causality

A system is causal if the output at time t depends only on the values of the input at times prior to or equal to t.

Note: *This is an example of a sidebar note block.*

3 Math Module: `stemset-math.sty`

This module provides optimized macros for frequent mathematical formulas in scientific contexts, reducing code verbosity.

3.1 What it does

It simplifies typing complex operators like derivatives, integrals, transforms (Fourier, Laplace, Z), and vector operators (gradient, divergence, curl), ensuring excellent spacing and typographical formatting.

3.2 How to Use (Code Example)

```
The Laplace transform of a derivative is:
\eqn{eq:laplace}{
  \laplace{\der{f(t)}{t}} = s \laplace{f(t)} - f(0)
}

Calculating curl and divergence:
\eqns{eq:vector}{
  \curl \mathbf{E} &= -\pder{\mathbf{B}}{t} \\
  \dive \mathbf{B} &= 0
}
```

3.3 Full List of Features

Below are the math macros available in the Paper template:

- **Equation Systems**: `\system{ x = 1 \\y = 2 }`
- **First Derivative**: `\der{f}{t}`
- **Second Derivative**: `\dder{f}{t}`
- **N-th Derivative**: `\nder{n}{f}{t}`
- **Partial Derivative**: `\pder{f}{x}`
- **Definite Integral**: `\intab{a}{b}{f(x)}{x}`
- **Limit Evaluation**: `\eval{x^2}{0}{1}`
- **Fourier Transform**: `\fourier{x(t)}`
- **Laplace Transform**: `\laplace{f(t)}`
- **Z Transform**: `\ztrans{x[n]}`
- **Phasor**: `\phasor{V}{\phi}` or `\fasore`
- **Real and Imaginary Part**: `\real{z}`, `\imag{z}`
- **Conjugate and Negation**: `\conj{z}`, `\notlog{A}`
- **Number Sets**: `\RR, \NN, \ZZ, \CC, \QQ`
- **Norm and Absolute Value**: `\norm{v}`, `\abs{x}`
- **Inner Product**: `\inner{u}{v}`
- **Vector Operators**: `\grad` (∇), `\dive` ($\nabla\cdot$), `\curl` ($\nabla\times$), `\lap` (∇^2)
- **Matrix Operations**: `\tr{A}`, `\det{A}`, `\rank{A}`, `\adj{A}`, `\matrixthree{1}...{9}`
- **Probability and Statistics**: `\prob{A}`, `\expval{X}`, `\var{X}`, `\cov{X}{Y}`
- **Limits, Sums, Products**: `\limit{x}{0}`, `\summ{i}{N}`, `\prodd{i}{N}`
- **Numbered Equations**: `\eqn{label}{...}` and for systems `\eqns{label}{...}`

3.4 Visual Result

The Laplace transform of a derivative is:

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\mathcal{L}\left\{f(t)\right\} - f(0) \tag{1}$$

Calculating curl and divergence:

$$\begin{aligned} \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \qquad (2) \\ \nabla \cdot \mathbf{B} &= 0 \qquad (3) \end{aligned}$$

4 Code Module: `stemset-code.sty`

Inserting code snippets in a paper requires excellent readability and correct syntax highlighting. The `stemset-code` module leverages `listings` to provide pre-configured code blocks.

4.1 What it does

It allows you to insert source code formatted with professional color schemes. Optimized pre-configured languages include: VHDL (Vivado style colors), C/C++, Python, Java, MATLAB, NASM, and Bash scripts. It also supports inline syntax for referencing variables or functions in plain text.

4.2 How to Use (Code Example)

```
An example of Python code (use \pythoninline{import numpy as np} in the text):

\begin{lstlisting}[style=python]
def calculate_mean(values):
    # Returns the arithmetic mean
    return sum(values) / len(values)
\end{lstlisting}
```

4.3 Full List of Features

Below are all the supported languages and their inline commands:

- **VHDL**: `\begin{lstlisting}[style=vhdl]` | Compact: `[style=vhdlcompact]` | Inline: `\vhdlinline{...}`
- **C**: `\begin{lstlisting}[style=c]` | Compact: `[style=ccompact]` | Inline: `\cinline{...}`
- **C++**: `\begin{lstlisting}[style=cpp]` | Compact: `[style=cppcompact]` | Inline: `\cppinline{...}`
- **Python**: `\begin{lstlisting}[style=python]` | Compact: `[style=pythoncompact]` | Inline: `\pythoninline{...}`
- **Java**: `\begin{lstlisting}[style=java]` | Compact: `[style=javacompact]` | Inline: `\javainline{...}`
- **MATLAB**: `\begin{lstlisting}[style=matlab]` | Compact: `[style=matlabcompact]` | Inline: `\matlabinline{...}`
- **LaTeX**: `\begin{lstlisting}[style=latex]` | Compact: `[style=latexcompact]`
- **NASM**: `\begin{lstlisting}[style=nasm]` | Compact: `[style=nasmcompact]` | Inline: `\nasminline{...}`
- **Bash**: `\begin{lstlisting}[style=bash]` (No compact or inline version)

4.4 Visual Result

An example of Python code (use `import numpy as np` in the text):

```python
def calculate_mean(values):
    # Returns the arithmetic mean
    return sum(values) / len(values)
```

```vhdl
entity D_FlipFlop is
    port ( clk, d : in std_logic; q : out std_logic );
end entity;
```

Listing 1: *D Flip Flop declaration in VHDL*

5 Algorithms Module: `stemset-algorithms.sty`

When describing the logical architecture of software, pseudocode is the most formal and elegant way. The `stemset-algorithms` module wraps the powerful `algorithm2e` package.

5.1 What it does

It allows you to lay out structured pseudocode blocks, complete with automatic numbering. The advantage of StemSet is instant localization: if you set `language=english` in `stemset.sty`, all keywords (*Input, Output, if, then, while*) are printed in English.

5.2 How to Use (Code Example)

```
\begin{algorithm}[H]
  \KwIn{An array $A$ of $n$ numbers}
  \KwOut{The maximum value}
  $max \gets A[0]$\;
  \For{$i \gets 1$ \KwTo $n-1$}{
    \If{$A[i] > max$}{
      $max \gets A[i]$\;
    }
  }
  \Return{$max$}\;
  \caption{Maximum Search}
\end{algorithm}
```

5.3 Full List of Features

The package automatically translates the standard commands of `algorithm2e`:

- **Input and Output**: `\KwIn{...}`, `\KwOut{...}`
- **For Loop**: `\For{condition}{...}`
- **While Loop**: `\While{condition}{...}`
- **If-Else Construct**: `\If{...}{...}`, `\ElseIf{...}{...}`, `\Else{...}`
- **Value Return**: `\Return{...}`
- **Assignment**: Use `\gets` to print the symbol $\leftarrow$

5.4 Visual Result

Algorithm 1: Maximum Search

Input: An array A of n numbers
Output: The maximum value
$max \leftarrow A[0]$;
for $i \leftarrow 1$ **to** $n-1$ **do**
  if $A[i] > max$ **then**
    $max \leftarrow A[i]$;
return max;

6 TikZ Module: `stemset-tikz.sty`

Writing pure TikZ code to create simple block diagrams or hardware datapaths is tedious. The `stemset-tikz` module provides high-level semantic commands to simplify the creation of consistent graphic nodes.

6.1 What it does

It defines TikZ macros that eliminate the need to write `\node[...]` code manually for control system block diagrams, hardware datapaths, and algorithm flowcharts.

6.2 How to Use (Code Example)

All macros share the same three-parameter signature: `\command{id}{text}{positioning}`

- `id`: unique identifier for the node, used to connect it to other elements (e.g. `alu1`).
- `text`: label displayed inside the block.
- `positioning`: TikZ relative coordinate (e.g. `right=of filter`, `below=1.5cm of start`). Leave this parameter empty for the first node (`{}`).

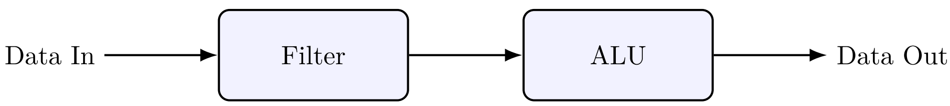
```
\begin{tikzpicture}[node distance=1.5cm, arrow/.style={-Latex, thick}]
  \node (input) {Data In};
  \sysblock{filter}{Filter}{right=of input};
  \sysblock{alu}{ALU}{right=of filter};
  \node[right=1.5cm of alu] (output) {Data Out};

  \draw[arrow] (input) -- (filter);
  \draw[arrow] (filter) -- (alu);
  \draw[arrow] (alu) -- (output);
\end{tikzpicture}
```

6.3 Full List of Features

Available commands are grouped by type:

- **Start/End Block (Red rounded rectangle)**: `\startendblock{id}{text}{pos}`
- **System Block (Blue rounded rectangle)**: `\sysblock{id}{text}{pos}`
- **Process Block (Orange rectangle)**: `\flowblock{id}{text}{pos}`
- **Decision Block (Green diamond)**: `\decblock{id}{text}{pos}`
- **Generic Block (Simple rectangle)**: `\genericblock{id}{text}{pos}`
- **Summing Junction (Circle with Sigma)**: `\sumnode{id}{pos}`
- **Gain Block (Gain triangle)**: `\gainblock{id}{text}{pos}`

6.4 Visual Result



7 Quick Reference and Cheat Sheet

7.1 Module Summary

Module: `math`

Code	Function	Use Case
`\system{x=1 \\y=2}`	$\begin{cases} x=1 \\ y=2 \end{cases}$	Equation systems
`\der{f}{t}`	$\frac{df}{dt}$	Ordinary derivatives
`\dder{f}{t}`	$\frac{d^2 f}{dt^2}$	Second derivatives
`\nder{n}{f}{t}`	$\frac{d^t n}{df^t}$	Higher-order derivatives
`\pder{f}{x}`	$\frac{\partial f}{\partial x}$	Partial derivatives
`\intab{a}{b}{f(x)}{x}`	$\int_a^b f(x)\,dx$	Definite integrals
`\eval{x^2}{0}{1}`	$\left[x^2\right]_0^1$	Limits of integration
`\limit{x}{0}`	$\lim_{x\to 0}$	Calculus limits
`\summ{i}{N}`	$\sum_{i=1}^N$	Series summation
`\prodd{i}{N}`	$\prod_{i=1}^N$	Product operators
`\fourier{x(t)}`	$\mathcal{F}\left\{x(t)\right\}$	Signal Fourier transforms
`\laplace{f(t)}`	$\mathcal{L}\left\{f(t)\right\}$	Control Laplace transforms
`\ztrans{x[n]}`	$\mathcal{Z}\left\{x[n]\right\}$	Discrete Z transforms
`\fasore{V}{\phi}`	$V\angle\phi$	AC circuits calculations
`\real{z}, \imag{z}`	$\Re\left\{z\right\},\Im\left\{z\right\}$	Complex analysis
`\conj{z}`	z^*	Complex analysis
`\notlog{A \cdot B}`	$\overline{A\cdot B}$	Digital electronics
`\RR, \NN, \ZZ, \CC`	$\mathbb{R},\mathbb{N},\mathbb{Z},\mathbb{C}$	Common number sets
`\norm{v}, \abs{x}`	$\left\lVert v\right\rVert,\left\lvert x\right\rvert$	Algebra & calculus
`\inner{u}{v}`	$\langle u,v\rangle$	Vector inner product
`\grad f, \dive \mathbf{v}, \curl \mathbf{v}`	$\nabla f,\nabla\cdot\mathbf{v},\nabla\times\mathbf{v}$	Vector calculus
`\lap f`	$\nabla^2 f$	Laplacian wave eq.
`\tr{A}, \det{A}`	$\mathrm{tr}(A),\det(A)$	Linear algebra
`\rank{A}, \adj{A}`	$\operatorname{rank} A,\operatorname{adj}(A)$	Linear algebra
`\matrixthree{1}..{9}`	$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$	Matrices insertion
`\prob{A}, \expval{X}`	$\Pr\left\{A\right\},\mathbb{E}!\left[X\right]$	Probability
`\var{X}, \cov{X}{Y}`	$\mathrm{Var}(X),\mathrm{Cov}(X,Y)$	Statistics
`\eqn{lbl}{..}`	Eq. (1)	Numbered equation
`\eqns{lbl}{..}`	Eq. (1), (2)	Aligned equations

Module: `boxes`

Code	Function	Use Case
`\definition` / `\definizione`	Definition Box	Core concept
`\theorem` / `\teorema`	Theorem Box	Mathematical rules
`\corollary` / `\corollario`	Corollary Box	Direct consequences
`\proposition` / `\proposizione`	Proposition Box	Minor statements
`\note` / `\nota`	Note Box	Sidebar annotations
`\warning` / `\attenzione`	Warning Box	Cautions and mistakes
`\example` / `\esempio`	Example Box	Practical examples
`\remember` / `\ricorda`	Remember Box	Important summaries
`\tip` / `\suggerimento`	Tip Box	Tricks and advice
`\examquestion` / `\domandaesame`	Exam Question Box	Study annotations
`\curiosity` / `\curiosita`	Curiosity Box	Historical trivia

Modules: `code` & `algorithms`

Code	Function	Use Case
\begin{lstlisting} [style=python]	Python Block	Analysis scripts
\begin{lstlisting} [style=vhdl]	VHDL Block	FPGA coding
\begin{lstlisting} [style=c]	C Block	Embedded systems
\begin{lstlisting} [style=cpp]	C++ Block	Object-oriented code
\begin{lstlisting} [style=java]	Java Block	Software design
\begin{lstlisting} [style=matlab]	MATLAB Block	Data & control systems
\begin{lstlisting} [style=bash]	Bash Block	Scripting/DevOps
\begin{lstlisting} [style=latex]	LaTeX Block	Technical manuals
\begin{lstlisting} [style=*lang*compact]	Compact code	Long listings (e.g. `ccompact`)
*lang*inline{code}	Inline code	Variables in text
\begin{algorithm}	Pseudocode block	Abstract logic
\KwIn{...}	Input / Output	Algorithm setup
\For{cond}{...}	FOR Loop	Known iterations
\While{cond}{...}	WHILE Loop	Unknown iterations
\If{...}{...}	IF / ELSE	Logical conditions
\Return{...}	Output	Terminate function

Module: `tikz`

Code	Function	Use Case
\startendblock{id}{text}{pos}	Start/end block	Flowcharts (red)
\sysblock{id}{text}{pos}	System block	Hardware/registers (blue)
\flowblock{id}{text}{pos}	Process block	Software tasks (orange)
\decblock{id}{text}{pos}	Decision block	Conditional branches (green)
\genericblock{id}{text}{pos}	Generic block	Simple rectangle
\sumnode{id}{pos}	Summing node	Signal summation
\gainblock{id}{text}{pos}	Gain block	Controller gain

7.2 Visual Cheat Sheet: Box Styles

Below is the visual appearance of all boxes and theorems provided by the template.

Definition: Definition

Example of a definition box.

Theorem 7.1. *TheoremExample of a theorem box.*

Note: *Example of a note box.*

Warning: *Example of a warning box.*

Example 7.0 – Example

Example of a numbered example box.

Remember

Remember this information.

Tip

A very useful tip for you.

Exam Question

This will definitely be in the exam.

Fun Fact

Did you know that...